

071306T4EEN

Electrical Engineering Level 6

ENG/OS/PO/CR/03/6

Install Electrical Machine

March / April 2023



THE KENYA NATIONAL EXAMINATIONS COUNCIL

WRITTEN ASSESSMENT GUIDE

INSTRUCTIONS TO ASSESSOR

Marks for each question are indicated in the brackets

*The paper consists of **TWO** sections: **A and B***

This paper consists of eleven (11) printed pages

Assessor should check the tool to ascertain that all the pages are printed as indicated and that no question is missing

SECTION A (40 MARKS)

These only serves as a guide to expected responses.

1. State any **four** importance of testing electrical machine installations (4 marks)
 - i. **Electrical faults and overloads can cause fires, explosions, and arcs.**
 - ii. **Electrical equipment can deteriorate, be damaged and malfunction and danger can arise when faults develop or power fails.**
 - iii. **Electrical equipment can be designed or installed incorrectly and create danger.**
 - iv. **The need for replacement, rewiring, or refurbishment can be identified and planned.**
 - v. **Unplanned and emergency repairs or replacement can be minimized.**
 - vi. **Condition based maintenance programs can be developed based on evidence.**

(award 1 mark for each correct X4)
2. State **four** requirements for mounting electrical machines (4 marks)
 - i. **The environment impacts the kind of machine that needs to be installed is assessed in order to secure the machine, operators and other equipment in the environments**
 - ii. **Ensure compliant with IEE regulations during mounting**
 - iii. **If the nameplate of the machine is hidden or unclear after installation, a new one must be put up someplace where it can easily be seen.**
 - iv. **Special consideration has to be given to the insulation**
 - v. **Clear signs and arrows should be propped up so that no reverse-rotation scenarios occur, causing health hazard or equipment damage**
 - vi. **Machines must be earthed with a wire made of copper, with one end in control panel that has machine starter.**
 - vii. **Boxes that contain the machine junction should be made of metal**
 - viii. **A motor must have a service factor of at least 1.15 if it is to work continuously or in an environment where the temperature exceeds 40C**

(award 1 mark for any for X4, accept any other correct)
3. State any **four** importance of earthing electrical machines (4 marks)
 - i. **To protect human lives as well as provide safety to machines and appliances from leakage current.**
 - ii. **To keep voltage as constant in the healthy phase (If fault occurs on any one phase).**
 - iii. **To Protect Electric system and buildings form lighting.**
 - iv. **To serve as a return conductor in electric traction system and control**

v. To avoid the risk of fire in machine installation systems.

(award 1 mark for any four X4)

4. State the THREE classes of single phase motors (3 marks)

- i. Synchronous motors**
- ii. Induction motors**
- iii. Universal motors**

(Award 3×1 marks)

5. State any TWO IEEE regulations governing electrical machine installation (2 marks)

- i. Every machine must have means of control**
- ii. Every machine must have means of protection**
- iii. Every machine must be wired using appropriate cables sizes**

(Award any 2 × 1)

6. State any **four** factors that need to be considered when installing motors for machine drives

(4 marks)

- i. Power output required from the motor**
- ii. Operating speed of the motor and the speed characteristics**
- iii. Speed control**
- iv. Method of starting**
- v. Starting torque**
- vi. Type of frame enclosure**
- vii. Nature of atmosphere and temperature in which it is to operate**
- viii. Nature of electrical power supply, that is single, three phase or d.c**

(Award 1 × 4 marks)

7. State any **three** merits of three phase over single-phase induction motors (3 marks)

- i. More efficient**
- ii. Cheap in terms of cost**
- iii. Smaller in size for the same rating**
- iv. Self-starting and better operational characteristics**

(Award any 3 × 1 mark for any correct answer)

8. Explain the meaning of continuous rating of an electric motor (5 marks)

As the machine runs, there is inevitable loss of power in form of heat. *(Award 1or 0 mark)*

The heat so generated accumulates with time and may rise to dangerous levels. (Award 1 or 0 mark)

Some machines are so designed to allow heat loss to the surrounding through convention. (Award 1 or 0 mark)

When heat gained = heat lost, the motor temperature does not rise again. (Award 1 or 0 mark)

Hence the motor continuously runs (Award 1 or 0 mark)

Such a motor is said to be rated continuous.

(Award 2 or 0 mark for any correct definition)

9. Define an electrical machine (2 Marks)

It is a device which converts mechanical energy into electrical energy or vice versa.

(Award 2 or 0 marks for definition)

10. Outline steps of aligning motor and load during installation (5 marks)

- **Prepare the foundation to be firm and level**
- **Mount the load firmly on the foundation**
- **By use of appropriate tool, align the motor and the load**
- **Mark out the mounting spots for the motor**
- **Fasten the motor on the foundation**

(Award 5 × 1)

11. State the FOUR methods of starting squirrel cage induction motors (4 marks)

- i. **Direct online (D.O.L) starting**(Award 1 or 0)
- ii. **Stator resistance starting**(Award 1 or 0)
- iii. **Autotransformer starting**(Award 1 or 0)
- iv. **Star-delta starting** (Award 1 or 0)

SECTION B (60 MARKS)

This only serves as a guide

12. a) Briefly discuss **four** factors you will consider when laying machine cables (8 marks)
- i. **Installation requirement-** Type of installation would require use of specific cables, Industrial machine installation involving machines would require armored cables, stronger fittings, VFD
 - ii. **Voltage rating of the Installation-** Calculated rating of cables required for machine installation is imperative to avoid damaging the equipment by short circuit faults
 - iii. **Type of power supply-** D.C cables should be used for D.C power supply while AC cables should also be applied where the power supply is AC.
 - iv. **Installation requirements-** Type of equipment involved in the installation would determine the type of cables used for installation. Control cables, communication cables, main circuit cables are laid according to installation requirements.
 - v. **Cable shielding-** When laying cables, it is important to consider type of insulation on the cable, whether the system is grounded and safety and reliability of the system

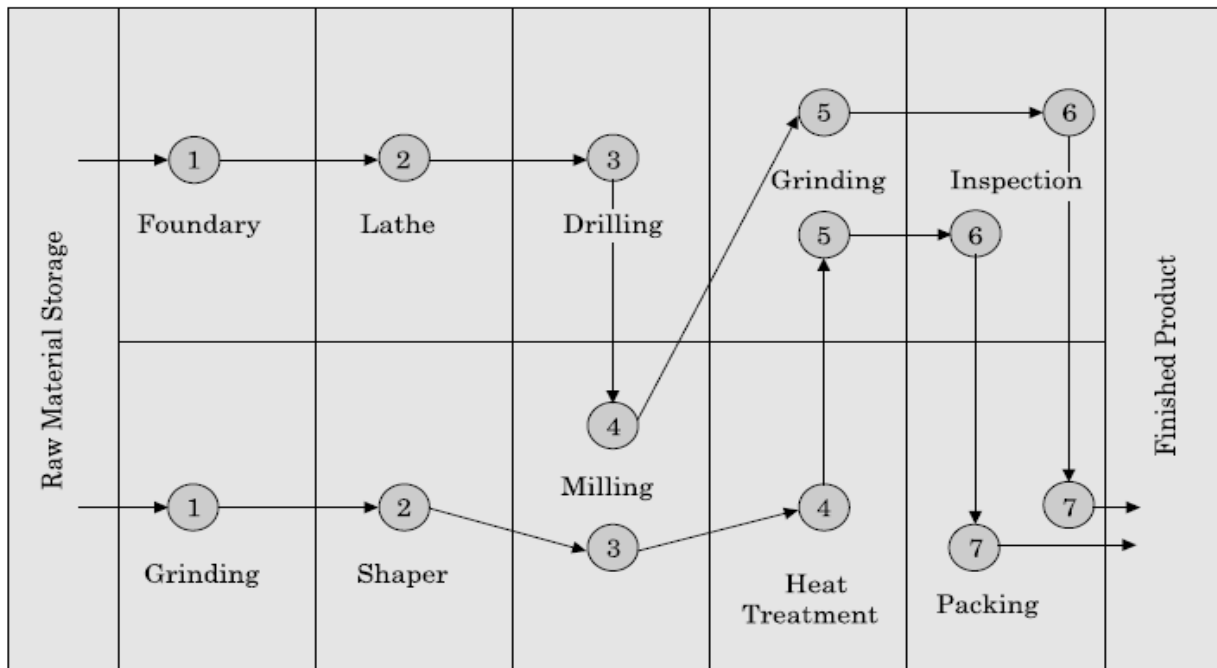
(1 mark for correct factor and 1 mark for correct explanation X4)

- b) With the aid of sketches and appropriate example in each case, describe the following types of machine layouts:

- i. Process layout (6 marks)

Process Layout

A process layout is the arrangement of facilities and equipment in group according to the function performed. Similar types of machines or similar types of operations are grouped together and put in one department. For example, all grinding machines may be at one place, all drilling machines at another place and all turning machines at yet another place. It means that all the machines have been arranged according to their functions. Examples of such layouts in manufacturing are machine shop, shoe making industries, garment industries, etc. and in service sector hospitals, universities, bank, etc. are the examples. They are organised so that their resources are grouped according to the different functions performed.



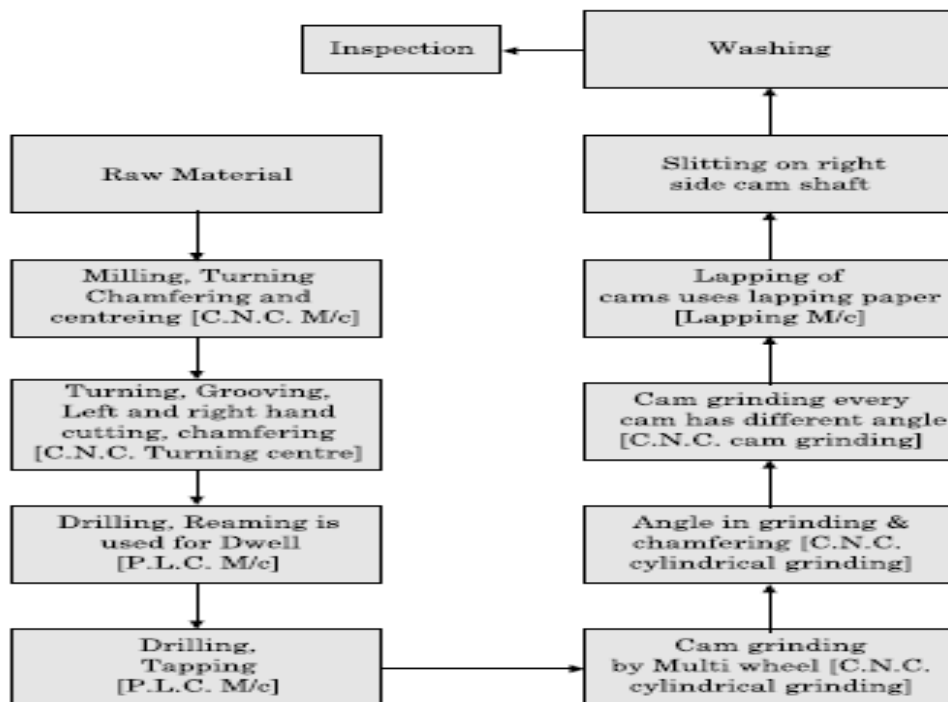
(award 2 marks for description , 1 mark for example and 3 marks for correct sketch)

ii. Product layout

(6 marks)

A product layout is the arrangement of facilities and equipment in the same sequence as that of the operations needed to complete each unit of the product or service offered.

Successive units follow the same path through the system. This type of layout is very popular in mass production, where only one type of product is manufactured. Examples of product layout applications in the manufacturing are automobile parts industries, chemical industries, T.V. set manufacturing, refrigeration manufacturing, bottling plants, etc. As the facilities in product layout are arranged in line as per the processing sequence, it is also referred to as line layout.



(award 2 marks for description , 1 mark for example and 3 marks for correct sketch)

13. a) Discuss any **four** tools and equipment applied in testing parameters on an electrical machines installation (8 marks)

- i. **Battery tester-** This is a type of electronic device that is created for the purpose and intent of testing a battery state. Most all test for charger within the cells of the battery and the overall voltage output. There are many that will actually test the condition of the battery, like its ability to accumulate a charge and any flaws that may detrimentally impact its overall performance.
- ii. **Multimeter-** This electronic instrument that measures electrical voltage, electrical current, and electrical resistance. These are often referred to as a “VOM Meter”. These electrical tools and equipment pieces may be used to troubleshoot electrical issues in equipment, controls associated with motors, appliances, wiring systems, and even power supplies.
- iii. **Outlet testers-** This tool is used to determine if a wall outlet that operates and/or distributes AC power is wired as it is supposed to be wired. In addition to this, it has the ability to determine if the outlet being tested has power.

iv. **Voltage tester**-It is commonly used for voltage tester aids in determining if electrical current is flowing properly through the wire being examined. Additionally, it determines if there is an accurate amount of voltage within the wire and if the wire is properly grounded.

v. **Wire tracer**-A wire tracer is a type of testing tool that has the ability to determine if there is a wire or cable that is unshielded or unenergized within a particular set-up

(award 1 mark for tool/equipment and one mark for discussion X4, accept any other correct)

b) Explain how the direction of rotation of the following motors can be reversed using diagrams:

v. Three phase induction motor

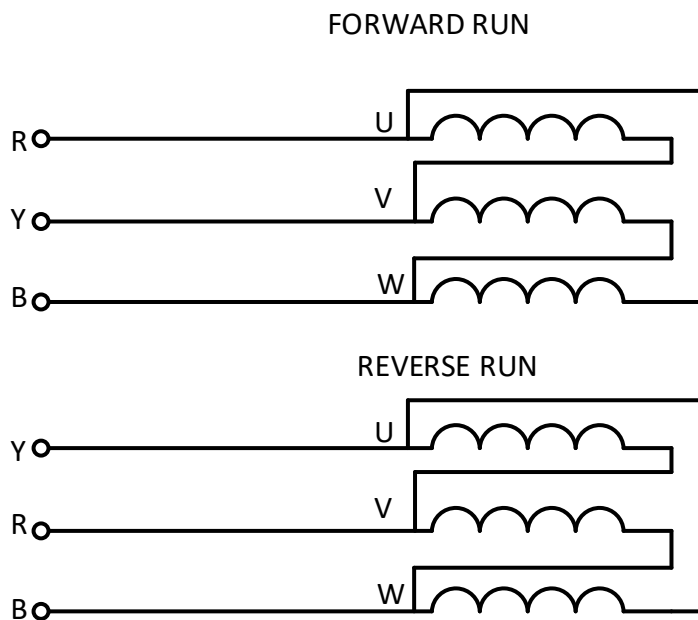
vi. Single phase capacitor start motor

vii. D.C shunt motor

viii. Universal motor

(12 marks)

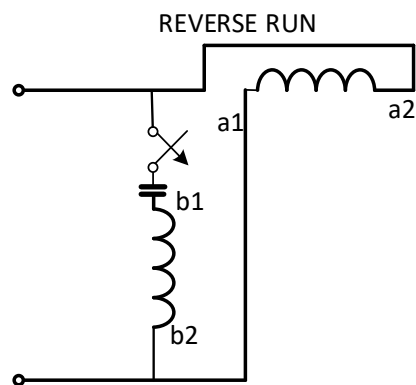
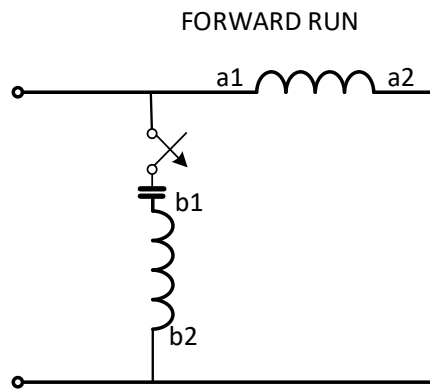
i. Three phase induction motor-**change any two supply leads**



(Award 2 or 0 for diagram)

(Award 1 or 0 for statement)

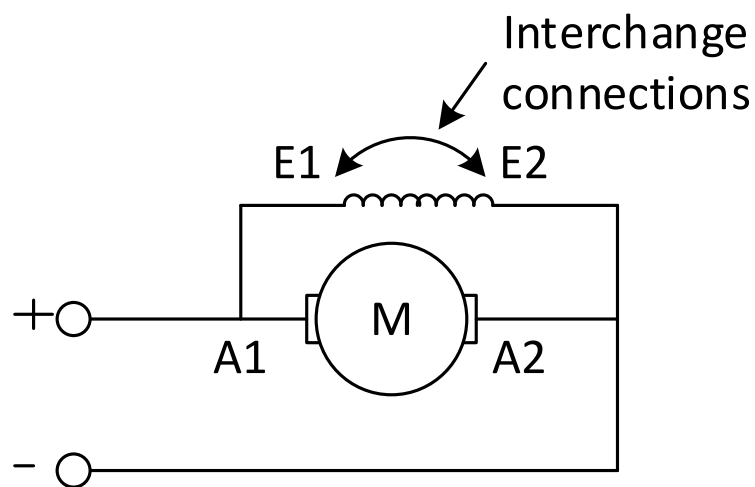
ii. Single phase capacitor start motor- **change connections of running winding or starting winding**



(Award 2 or 0 for diagram)

(Award 1 or 0 for statement)

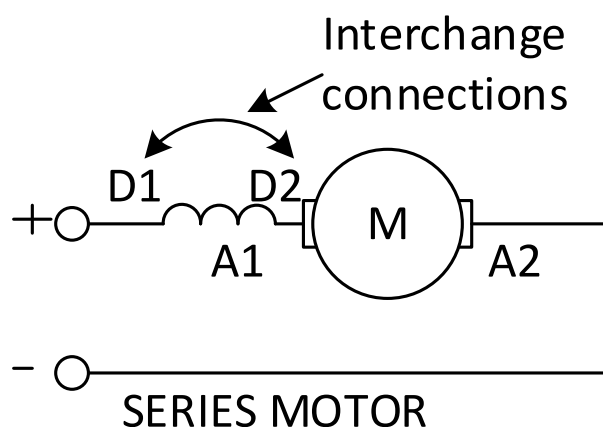
iii. D.C shunt motor- **change connections of shunt field winding or armature winding**



(Award 2 or 0 for diagram)

(Award 1 or 0 for statement)

iv. Universal motor- **change connections of series field winding or armature winding**



(Award 2 or 0 for diagram)

(Award 1 or 0 for statement)

14. a) Explain any **four** safety requirements during mounting of electrical machines (8 marks)
- i. **Develop, document, implement and maintain appropriate work procedures, measures, inspections, and precautions and emergency procedures to control the hazards that may be present by using these guidelines and in accordance to applicable legislation and standards.**
 - ii. **Ensure all written procedures are readily available to machine users**
 - iii. **Ensure that the machines and power tools are maintained and in good condition.**
 - iv. **Establish machine-specific Lockout and Tag out procedures.**
 - v. **Ensure that all persons operating the equipment are provided with and wear the appropriate personal protective equipment (PPE).**
 - vi. **Participate in appropriate training to work safely with machines and power tools**
 - vii. **Review and be familiar with applicable work procedures before start of work.**
 - viii. **Operate the machines and power tools according to work procedures and manufacturer instructions**
 - ix. **Inspect any equipment for defects prior to its use**
- (award 2 marks for each explanationX4)*

b) A 440V ,3 phase 50Hz, 37.3kW, Y connected induction motor has the following parameters

$$R_1 = 0.1\Omega, X_1 = 0.4\Omega, R_2' = 0.15\Omega, X_2' = 0.15\Omega$$

The motor has stator core loss of 1250W and rotational loss of 1000W. It draws a no load line current of 20A at a pf of 0.09(lag). When the motor operates at a slip of 3%, calculate:

- i. Input line current

$$I_1 = I_0 + I_2' \text{ (Award 1 mark)}$$

$$\text{Applied voltage per phase, } V_1 = \frac{V_1}{\sqrt{3}} = \frac{400}{\sqrt{3}} = 230.94 \angle 0^\circ$$

$$\begin{aligned} I_1/2 &= V_1 / (R_1 + R_2'/s) + j(X_1 + X_2'/2) \\ &= 230.94 \angle 0^\circ / (0.1 + 0.15/0.03) + j(0.4 + 0.44) \\ &= 49.1 \angle -9.3^\circ \text{ (Award 1 mark)} \end{aligned}$$

For all practical purpose, no load motor current may be taken as equal to magnetizing current I_0

$$\begin{aligned} I_0 &= 20 \angle -84.9^\circ \\ &= 1.78 - j19.9 \text{ (Award 1 mark)} \end{aligned}$$

$$\begin{aligned} I_1 &= I_0 + I_1/2 = 1.78 - j19.9 + 48.4 - j7.9 \\ &= 50.2 - j27.8 \\ &= 57.4 \angle -29^\circ \text{ (Award 1 mark)} \end{aligned}$$

$$\text{Pf} = \cos 29^\circ = 0.875(\text{lag})$$

ii. Electromagnet torque developed in N-M

$$\text{Power transferred to rotor} = 3I_1/2 R_2'/s \text{ (Award 1 mark)}$$

$$\begin{aligned} P_2 &= 3 \times 49.1^2 \times 0.15 / 0.03 \\ &= 36160 \text{ W} \end{aligned}$$

$$\begin{aligned} \text{Torque, } T_g &= 9.55 \times 3I_1/2 R_2'/s \text{ (Award 1 mark)} \\ &= 9.55 \times 3 \times 36160 / 1500 \\ &= 230 \text{ N-M (Award 1 mark)} \end{aligned}$$

iii. Output power

$$\begin{aligned} \text{Mechanical power developed by rotor} &= 3I_1/2 R_2'(1-s/s) \\ &= (1-s) P_2 \text{ (Award 1 mark)} \\ &= (1 - 0.03)36160 \\ &= 35073 \text{ W} \end{aligned}$$

$$\text{Output power} = 35073 - 1000 = 34073 \text{ W (Award 1 mark)}$$

iv. Efficiency of the motor

$$\begin{aligned} \text{Stator cu loss} &= 3(I_1/2)^2 R_1 \\ &= 3 \times 57.4^2 \times 0.1 \\ &= 988 \text{ W (Award 1 mark)} \\ \text{Rotor cu loss} &= 3(I_1/2)^2 R_2' = sP_2 \end{aligned}$$

$$= 0.03 \times 36160$$

$$= 1085\text{W (Award 1 mark)}$$

$$\text{Total losses} = 1250 + 988 + 1085 + 1000 = 4323\text{W}$$

$$\text{Efficiency} = \frac{34075}{34075 + 4323} = 0.887 \text{ or } 88.7\% \text{ (Award 1 mark)} \quad (12 \text{ marks})$$

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